Connectivity-Preserving V-Formation Control for Unicycle Multi-Robot Systems via Feedback Linearization and Adaptive Artificial Potential Fields

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***Abstract*—This paper addresses decentralized formation control for a group of five unicycle-type mobile robots navigating a cluttered two-dimensional environment while simultaneously preserving inter-agent communication connectivity. We extend the unified artificial potential field (APF) framework of Kan *et al.* to accommodate nonholonomic unicycle kinematics through feedback linearization via a look-ahead point transformation, which maps each robot onto a fully actuated virtual point and enables direct application of gradient-descent control laws. Four novel mechanisms augment the baseline APF architecture: (1) a *dynamic formation squeeze* that adaptively relaxes the rigid V-shape geometry when navigating narrow corridors; (2) a *saturated tangent barrier* for connectivity preservation that applies zero force in the safe regime and grows to a hard-capped restoring magnitude only when an inter-agent link approaches the communication threshold; (3) *distributed target tracking* without a single leader, enabling automatic network self-recovery after transient topological splits; and (4) a *non-conservative vortex field* superimposed on each obstacle's repulsion that breaks the rotational symmetry of the conservative APF landscape, eliminating saddle-point and local-minimum traps provably without affecting the Lyapunov energy-descent guarantee. A Lyapunov argument demonstrates that the safe, connected invariant set is forward invariant under the proposed control law. The five APF hyperparameters are tuned via Bayesian optimization over Gaussian processes [1] across a multi-environment gauntlet. Simulation results in both a structured staggered gate field and a random obstacle forest confirm successful goal acquisition, zero collisions, and strict maintenance of $\lambda_2(L(t)) > 0$ throughout all trials.**

I. Introduction

Coordinated multi-robot systems have attracted sustained research interest due to their transformative potential in industrial automation, warehouse logistics, and search-and-rescue operations [2]. A fundamental capability in such systems is *formation control*: the ability to drive a group of robots into a prescribed geometric configuration and maintain it while navigating a shared environment. Methods based on *Artificial Potential Fields* (APF) [3], [4], [5] are particularly attractive for their analytical tractability and real-time implementability, unifying formation, obstacle, and connectivity objectives in a single gradient-descent control law amenable to decentralized implementation.

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A critical requirement frequently overlooked in APF-based formation controllers is the preservation of inter-agent *communication connectivity*. If agents drift beyond each other's sensing range during obstacle-avoidance maneuvers, the communication graph becomes disconnected and cooperative control breaks down entirely. Despite the maturity of both APF-based formation control and nonholonomic motion planning as individual fields, no existing work provides a unified treatment of formation stabilization, obstacle avoidance, and connectivity preservation for unicycle-type multi-robot systems — the platform class predominant in warehouse and field deployments.

This paper addresses that gap with four contributions. The primary contribution extends the APF framework of [6] from point agents to unicycle robots via feedback linearization, with a Lyapunov stability proof. Three additional mechanisms resolve the practical failure modes of rigid APF architectures in highly cluttered environments: the *dynamic formation squeeze* for navigating narrow corridors, the *saturated tangent barrier* for zero-drag connectivity preservation, and a *per-obstacle vortex field* that eliminates local minima and saddle points provably without violating the energy-descent guarantee.

II. Related Work

APF-based formation control. Artificial Potential Fields, originally proposed by Khatib [3] for single-robot navigation, were extended to the multi-agent setting by Leonard and Fiorelli [4], who formulated inter-agent attractive and obstacle repulsive potentials that drive a fleet toward a desired geometric configuration. Rimon and Koditschek [7] established that such potentials can be constructed as provably complete *navigation functions* for sphere-world environments, guaranteeing convergence from almost all initial configurations. Ge and Cui [8] later addressed the ubiquitous local-minimum problem by redesigning the repulsive term to incorporate the robot-to-goal distance, eliminating spurious equilibria in cluttered scenes. Olfati-Saber [5] extended potential-field ideas to multi-agent flocking, providing a cohesion–separation–velocity-matching framework that became a canonical benchmark for decentralized multi-robot design.

Communication connectivity preservation. The algebraic connectivity $\lambda_2(L)$ of the graph Laplacian L provides a spectral certificate for network connectivity: the communication graph is connected if and only if $\lambda_2(L) > 0$ [9]. Ji and

Egerstedt [9] showed that connectivity maintenance can be formulated as a hard control constraint, while Dimarogonas and Kyriakopoulos [10] developed distributed controllers that guarantee $\lambda_2(L(t)) > 0$ for all time. Panagou *et al.* [11] introduced smooth barrier functions of λ_2 as additional potential terms, at the cost of a continuous connectivity drag on all inter-agent motion even in the open field.

Unified APF with connectivity for point agents. Kan *et al.* [6] represent the closest existing work, unifying formation stabilization, obstacle avoidance, and connectivity preservation within a single APF framework and proving convergence via a Lyapunov argument. That result is, however, derived exclusively for *point agents* governed by single-integrator dynamics $\dot{p}_i = u_i$, $u_i \in \mathbb{R}^2$, in which the velocity vector can be set to any direction instantaneously. The Pfaffian nonholonomic constraint of unicycle robots prohibits lateral velocity and fundamentally invalidates the gradient-descent controller of [6]. Do [12] addressed formation tracking for unicycle-type robots with limited sensing ranges, but without a connectivity-preservation mechanism. The present work closes this gap by combining all three objectives for the first time in the unicycle setting.

III. Problem Formulation

The system consists of $N = 5$ unicycle robots operating in a planar warehouse environment with static obstacles. We formalize the agent kinematics, environment, inter-agent communication topology, desired formation, potential function, and control objectives.

A. Agent Kinematics and State Space

Let $\mathcal{I} = \{1, \ldots, N\}$ index the robot fleet. Robot $i \in \mathcal{I}$ is modeled as a unicycle (differential-drive) agent with configuration $\mathbf{q}_i = [x_i,\, y_i,\, \theta_i]^\top \in \mathbb{R}^2 \times S^1$, where (x_i, y_i) is the Cartesian position and $\theta_i \in (-\pi, \pi]$ is the heading angle. The continuous-time kinematics are

$$\dot{x}_i = v_i \cos\theta_i, \quad \dot{y}_i = v_i \sin\theta_i, \quad \dot{\theta}_i = \omega_i, \tag{1}$$

where $v_i \in \mathbb{R}$ and $\omega_i \in \mathbb{R}$ are the linear and angular velocity inputs, respectively. Let $p_i = [x_i,\, y_i]^\top \in \mathbb{R}^2$ denote the planar position of agent i and define the stacked position vector $\mathbf{p} = \operatorname{stack}_{i\in\mathcal{I}} p_i \in \mathbb{R}^{2N}$.

The unicycle is subject to the Pfaffian nonholonomic constraint

$$-\dot{x}_i \sin\theta_i + \dot{y}_i \cos\theta_i = 0, \tag{2}$$

which confines the instantaneous velocity to the body-fixed longitudinal axis and prohibits lateral motion. Constraint Eq. (2) invalidates the gradient-descent control laws designed for single-integrator (point) agents in [6], since $[\dot{x}_i,\, \dot{y}_i]^\top$ cannot be set to an arbitrary direction independently of θ_i.

Assumption 1: The control inputs are saturated: $|v_i(t)| \le v_{\max} = 1.8$ m/s and $|\omega_i(t)| \le \omega_{\max} = 2.5$ rad/s for all $i \in \mathcal{I}$, $t \ge 0$.

B. Workspace and Obstacle Model

The robots operate within a convex, compact workspace $\mathcal{W} \subset \mathbb{R}^2$. The environment contains $M \ge 1$ static circular obstacles; obstacle $j \in \{1, \ldots, M\}$ is the closed disk

$$\mathcal{O}_j = \{p \in \mathbb{R}^2 : \|p - c_j\| \le r_j\}, \tag{3}$$

with center $c_j \in \mathbb{R}^2$ and radius $r_j > 0$. The obstacle-free region is

$$\mathcal{W}_{\text{free}} = \mathcal{W} \setminus \bigcup_{j=1}^{M} \mathcal{O}_j. \tag{4}$$

Assumption 2: The obstacles are known a priori, pairwise non-overlapping ($\mathcal{O}_j \cap \mathcal{O}_k = \emptyset$ for $j \ne k$), and strictly contained in the interior of $\mathcal{W}$. Furthermore, $\mathcal{W}_{\text{free}}$ is connected.

Assumption 3: All robots begin in collision-free configurations: $p_i(0) \in \mathcal{W}_{\text{free}}$ for all $i \in \mathcal{I}$.

C. Communication Graph and Algebraic Connectivity

Inter-agent communication is modeled by the time-varying undirected graph $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$, where $\mathcal{V} = \mathcal{I}$ and the edge set

$$\mathcal{E}(t) = \{(i,j) \in \mathcal{V}^2 : i \ne j,\ \|p_i(t) - p_j(t)\| \le d_{\text{th}}\} \tag{5}$$

is determined by the proximity threshold $d_{\text{th}} > 0$. The binary adjacency matrix $A(t) = [a_{ij}(t)] \in \{0,1\}^{N\times N}$ satisfies $a_{ij} = a_{ji} = 1$ when $(i,j) \in \mathcal{E}(t)$ and zero otherwise. The graph Laplacian is $L(t) = D(t) - A(t)$, where $D(t) = \operatorname{diag}\left(\sum_j a_{1j}, \ldots, \sum_j a_{Nj}\right)$.

Lemma 1 (Connectivity criterion [9]): The graph $\mathcal{G}(t)$ is connected if and only if $\lambda_2(L(t)) > 0$, where $\lambda_2(\cdot)$ denotes the second-smallest eigenvalue (algebraic connectivity) of the Laplacian.

Assumption 4: The initial communication graph is connected, i.e., $\lambda_2(L(0)) > 0$.

Maintaining $\lambda_2(L(t)) > 0$ for all $t \ge 0$ ensures that the network remains connected throughout the mission, a property studied extensively in [10], [9].

D. Formation Specification

The desired V-shape formation is encoded by a set of reference inter-agent displacement vectors $\delta^*_{ij} \in \mathbb{R}^2$, defined for each pair (i,j) in the formation edge set $\mathcal{E}_0 \subseteq \mathcal{V} \times \mathcal{V}$. Agent 1 occupies the tip; agents 2, 4 form the starboard wing; agents 3, 5 form the port wing. The full template matrix $\Delta = \{\delta^*_{ij}\}_{i,j\in\mathcal{I}}$ is pre-loaded onto every agent before deployment; no formation geometry is exchanged at runtime, preserving the fully decentralized character of the architecture. Each agent uses only its own row $\{\delta^*_{ij}\}_{j\in\mathcal{N}_i}$ and the relative positions of its current communication neighbors to compute its local formation error. The formation tracking error for pair $(i,j) \in \mathcal{E}_0$ is

$$e_{ij}(t) = \big(p_i(t) - p_j(t)\big) - \delta^*_{ij}, \tag{6}$$

and the global formation error vector is $e(t) = \operatorname{stack}_{(i,j)\in\mathcal{E}_0} e_{ij}(t) \in \mathbb{R}^{2|\mathcal{E}_0|}$. The formation is achieved when $\|e(t)\| \to 0$.

E. Unified Artificial Potential Function

Following the multi-agent APF paradigm of [4] and the connectivity-preserving framework of [6], we define a unified smooth potential function $U : \mathbb{R}^{2N} \to \mathbb{R}_{\geq 0}$ as

$$U(\mathbf{p}) = U_{\text{form}}(\mathbf{p}) + U_{\text{obs}}(\mathbf{p}) + U_{\text{conn}}(\mathbf{p}), \tag{7}$$

where each term targets a distinct control objective.

1) Formation potential (attractive):

$$U_{\text{form}}(\mathbf{p}) = \frac{1}{2} \sum_{(i,j)\in\mathcal{E}_0} \|e_{ij}\|^2. \tag{8}$$

2) Obstacle-avoidance potential (repulsive, Khatib [3]):

$$U_{\text{obs}}(\mathbf{p}) = \sum_{i\in\mathcal{I}} \sum_{j=1}^{M} \frac{k_{\text{rep}}}{2} \left(\frac{1}{d_{ij}} - \frac{1}{d_0} \right)^2 \mathbf{1}[d_{ij} < d_0], \tag{9}$$

where $d_{ij} = \|p_{v,i} - c_j\| - r_j$ is the surface-to-surface clearance between the look-ahead point of agent i and obstacle j, $d_0 > 0$ is the obstacle influence radius, and $k_{\text{rep}} > 0$ is a gain.

3) Connectivity-preservation potential (barrier):

$$U_{\text{conn}}(\mathbf{p}) = \gamma\big(\lambda_2(L(\mathbf{p}))\big), \tag{10}$$

where $\gamma : \mathbb{R}_{>0} \to \mathbb{R}_{\geq 0}$ is smooth, strictly decreasing, and satisfies $\gamma(\lambda_2) \to \infty$ as $\lambda_2 \to 0^+$, following the approach of [11].

F. Control Objective

Problem 1: Given $N = 5$ unicycle robots Eq. (1) with initial conditions $\mathbf{q}(0) \in \mathcal{W}_{\text{free}}^N$ and a connected initial graph satisfying Assumption 4, design a decentralized feedback control law

$$(v_i, \omega_i) = \kappa_i\big(\mathbf{q}_i,\ \{p_j : j \in \mathcal{N}_i(t)\}\big), \quad i \in \mathcal{I}, \tag{11}$$

where $\mathcal{N}_i(t) = \{j : (i,j) \in \mathcal{E}(t)\}$, such that:

1) **Formation convergence:** $\|e(t)\| \to 0$ as $t \to \infty$;
2) **Obstacle avoidance:** $p_i(t) \in \mathcal{W}_{\text{free}}$ for all $i \in \mathcal{I}$;
3) **Connectivity preservation:** $\lambda_2(L(t)) > 0$ for all $t \geq 0$.

IV. Proposed Controller Architecture

The proposed architecture overcomes the nonholonomic constraints of unicycle kinematics and the deadlock vulnerabilities of traditional APF methods [6]. The system is designed around a decentralized distributed tracking topology, enhanced by non-conservative vortex fields and adaptive formation mechanics.

A. Feedback Linearization via Look-Ahead Point

To bypass the Pfaffian constraint in Eq. (2), we employ the kinematic feedback linearization of [13] and define a virtual look-ahead point $p_{v,i} \in \mathbb{R}^2$ located at distance $l > 0$ along the longitudinal axis of robot i:

$$p_{v,i} = \begin{bmatrix} x_i + l\cos\theta_i \\ y_i + l\sin\theta_i \end{bmatrix}. \tag{12}$$

Differentiating Eq. (12) yields the decoupled kinematics:

$$\dot{p}_{v,i} = \underbrace{\begin{bmatrix} \cos\theta_i & -l\sin\theta_i \\ \sin\theta_i & l\cos\theta_i \end{bmatrix}}_{B(\theta_i)} \begin{bmatrix} v_i \\ \omega_i \end{bmatrix}. \tag{13}$$

Since $\det(B(\theta_i)) = l \neq 0$, the matrix is globally invertible. Setting the virtual control $u_{v,i} = \dot{p}_{v,i}$, the physical commands are exactly recovered as $(v_i, \omega_i)^\top = B(\theta_i)^{-1} u_{v,i}$. This maps the nonholonomic unicycles onto fully actuated single integrators at $p_{v,i}$, allowing direct application of gradient-based control laws. The look-ahead distance $l = 0.2$ m is chosen to balance decoupling fidelity against sensitivity to heading noise.

B. Distributed Target Tracking and Network Self-Recovery

To eliminate the single-point-of-failure inherent in strict leader-follower topologies, the global attractive potential is distributed across all agents. Each agent $i \in \mathcal{I}$ tracks a unique, offset spatial target:

$$p_{\text{target},i} = p_{\text{goal}} + \delta_{i,1}^*, \tag{14}$$

where $\delta_{i,1}^* = V_i - V_1$ is the desired displacement of agent i relative to agent 1 in the V-shape template V. Letting $\boldsymbol{\delta}_i = p_{v,i} - p_{\text{target},i}$, the normalized attractive force is

$$F_{\text{global},i} = \begin{cases} -k_{\text{att}} \dfrac{\boldsymbol{\delta}_i}{\|\boldsymbol{\delta}_i\|} & \|\boldsymbol{\delta}_i\| > 0.1 \text{ m}, \\ \mathbf{0} & \text{otherwise}. \end{cases} \tag{15}$$

The 0.1 m dead-zone suppresses chattering once an agent reaches its personal target. This architecture provides inherent *network self-recovery*: if extreme obstacle density transiently severs the communication graph, the disconnected sub-swarms independently advance toward their respective target coordinates and naturally reconverge, allowing the connectivity algorithm to automatically re-establish $\lambda_2 > 0$ without centralized intervention.

C. Adaptive Geometry: The Dynamic Formation Squeeze

In highly cluttered environments, rigid formation preservation structurally conflicts with obstacle avoidance. When a wide formation attempts to pass through a narrow gate, the outer agents are violently repelled, anchoring the swarm in a topological deadlock. To resolve this, each agent computes its minimum clearance to the nearest obstacle *and to all four map boundaries*:

$$d_{\min,i} = \min\left(\min_j \big(\|p_{v,i} - c_j\| - r_j\big),\ d_{\text{walls},i} \right), \tag{16}$$

where $d_{\text{walls},i}$ is the smallest signed distance to any boundary face. The formation gain is then scaled by

$$k_{\text{form},i}(t) = k_{\text{form}} \max\left(0.1,\ \frac{d_{\min,i}}{d_0} \right). \tag{17}$$

As $d_{\min,i} \to 0$, the drive to hold formation collapses to 10% of its baseline strength, allowing agents to squeeze through tight corridors. Once the swarm clears the obstacle zone ($d_{\min,i} \geq d_0$), the gain restores to 100% and the V-shape snaps back.

D. Saturated Tangent Barrier for Connectivity Preservation

Traditional rational barrier functions [11] apply continuous, distance-proportional drag on the network, severely degrading forward velocity even in the open field. We propose a *saturated tangent barrier* that acts as a slack constraint. For each connected pair $(i,j) \in \mathcal{E}(t)$, define the safe distance $d_{\text{safe}} = 0.85\, d_{\text{th}}$. Defining the inter-agent unit vector $\hat{n}_{ij} = (p_{v,i} - p_{v,j})/d_{ij}$ and the saturated barrier magnitude $\sigma_{ij} = \min\!\big(\tan(\frac{\pi}{2}\theta_{\text{d}}),\, F_{\text{cap}}\big)$, the connectivity restoring force is

$$F_{\text{conn},i} = \begin{cases} \mathbf{0} & d_{ij} \le d_{\text{safe}}, \\ -k_{\text{conn}}\, \sigma_{ij}\, \hat{n}_{ij} & d_{\text{safe}} < d_{ij} < d_{\text{th}}, \end{cases} \tag{18}$$

where $d_{ij} = \|p_{v,i} - p_{v,j}\|$, $\theta_{\text{d}} = (d_{ij} - d_{\text{safe}})/(d_{\text{th}} - d_{\text{safe}})$ is the normalized danger ratio clipped to $[0,\, 0.99]$, and $F_{\text{cap}} = 50.0$ N is the hard saturation cap. The cap prevents numerical blow-up near $d_{ij} \to d_{\text{th}}$ in discrete-time implementation, acting as a shock absorber while guaranteeing $\lambda_2 > 0$ in continuous time.

E. Repulsive Force and Non-Conservative Vortex Field

The repulsive force on agent i from obstacle j is the negative gradient of Eq. (9) with respect to $p_{v,i}$:

$$F_{\text{rep},ij} = k_{\text{rep}} \left(\frac{1}{d_{ij}} - \frac{1}{d_0} \right) \frac{\hat{n}^o_{ij}}{d^2_{ij}}, \tag{19}$$

where $\hat{n}^o_{ij} = (p_{v,i} - c_j)/\|p_{v,i} - c_j\|$ is the outward unit normal; active only when $d_{ij} < d_0$. Conservative APF landscapes are known to harbour saddle points and local minima that trap gradient-descent controllers [7], [8]. To escape these traps, a non-conservative *vortex field* is superimposed on each per-obstacle repulsion independently:

$$F_{\text{vortex},ij} = k_{\text{vortex}}\, R_{90}\, F_{\text{rep},ij}, \qquad R_{90} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \tag{20}$$

The total virtual control input combining all forces is

$$u_{v,i} = F_{\text{att},i} + \sum_j (F_{\text{rep},ij} + F_{\text{vortex},ij}) + F_{\text{conn},i}. \tag{21}$$

F. Penetration Fail-Safe

In discrete-time simulation, a robot's velocity can cause it to mathematically pierce a repulsive boundary before the continuous-time gradient reacts, driving $d_{ij} \to 0$ or below and making the APF force evaluate to infinity or NaN. A fail-safe overrides the standard gradient for any agent whose clearance drops below $d_{\text{pen}} = 0.05$ m:

$$F_{\text{rep},i} = \begin{cases} 50.0\, \hat{n}_{ij} & \text{if } d_{ij} \le 0.05, \\ \text{(normal APF)} & \text{otherwise}, \end{cases} \tag{22}$$

where $\hat{n}_{ij} = (p_{v,i} - c_j)/\|p_{v,i} - c_j\|$ is the outward unit normal. No vortex is applied in this regime; only a massive outward spring force ejects the agent.

V. Stability Analysis

We now prove that the proposed control architecture guarantees obstacle avoidance, connectivity preservation, and convergence to the desired formation, while the non-conservative vortex field neutralizes local minima without destroying the energy-descent guarantee.

Let the total conservative energy of the multi-agent system be defined by the scalar Lyapunov candidate:

$$V(\mathbf{p}_v) = U_{\text{form}}(\mathbf{p}_v) + U_{\text{obs}}(\mathbf{p}_v) + U_{\text{conn}}(\mathbf{p}_v). \tag{23}$$

By design, $V(\mathbf{p}_v) \ge 0$ and $V \to \infty$ at the boundaries of both $\mathcal{W}^N_{\text{free}}$ (due to U_{obs}) and the connected set $\{\lambda_2 > 0\}$ (due to U_{conn}).

The closed-loop dynamics for the virtual points are governed by the negative gradient of V supplemented by the vortex field:

$$\dot{p}_{v,i} = -\nabla_{p_{v,i}} V(\mathbf{p}_v) + F_{\text{vortex},i}, \tag{24}$$

where $F_{\text{vortex},i} = \sum_j F_{\text{vortex},ij}$ accumulates contributions from all active obstacles and walls.

Theorem 1: Under the control law Eq. (21) and the feedback linearization Eq. (13), the safe connected set $\Omega = \{\mathbf{p}_v \in \mathcal{W}^N_{\text{free}} : \lambda_2(L) > 0\}$ is forward invariant, and the system asymptotically converges to the desired formation.

Proof: Evaluating the orbital derivative of V along trajectories of Eq. (24) gives

$$\begin{aligned} \dot{V} &= \sum_{i=1}^{N} \big(\nabla_{p_{v,i}} V\big)^\top \dot{p}_{v,i} \\ &= -\sum_{i=1}^{N} \|\nabla_{p_{v,i}} V\|^2 + \sum_{i=1}^{N} \big(\nabla_{p_{v,i}} V\big)^\top F_{\text{vortex},i}. \end{aligned} \tag{25}$$

Because each vortex contribution is constructed as a 90° rotation of the corresponding repulsive gradient, $F_{\text{vortex},ij} = k_{\text{vortex}} R_{90} F_{\text{rep},ij}$, and $\nabla_{p_{v,i}} U_{\text{obs}} = \sum_j (-F_{\text{rep},ij})$, the cross-term for each obstacle satisfies

$$\begin{aligned} &\big(-F_{\text{rep},ij}\big)^\top \big(R_{90} F_{\text{rep},ij}\big) \\ &\quad = \big(F_{\text{rep},ij}\big)^\top \big(-R_{90} F_{\text{rep},ij}\big) \equiv 0, \end{aligned} \tag{26}$$

since the dot product of any vector with its orthogonal complement is identically zero. Therefore the vortex field performs strictly zero work on the conservative energy, and

$$\dot{V}(\mathbf{p}_v) = -\sum_{i=1}^{N} \|\nabla_{p_{v,i}} V\|^2 \le 0. \tag{27}$$

Since V is positive definite and radially unbounded on Ω, and $\dot{V} \le 0$, the trajectories can never escape Ω. By LaSalle's Invariance Principle, the system converges to the largest invariant subset of $\{\dot{V} = 0\} = \{\nabla_{p_{v,i}} V = 0\ \forall i\}$, which corresponds to the desired formation and goal state. ■

Remark 1: The Lyapunov argument above is stated for continuous-time dynamics. The discrete-time implementation with $dt = 0.05$ s introduces a residual numerical error; the penetration fail-safe Eq. (22) and the barrier saturation cap F_{cap} together bound this error and preserve the invariance of Ω in practice.

VI. Simulation and Bayesian Optimization

A. Simulation Environment

The controller was deployed in a discrete-time simulation using the parameters summarized in Table I.

TABLE I
SIMULATION AND CONTROLLER PARAMETERS

Parameter	Symbol	Value
Timestep	dt	0.05 s
Max simulation time	$t_{\max}$	90 s
Linear velocity limit	$v_{\max}$	1.8 m/s
Angular velocity limit	$\omega_{\max}$	2.5 rad/s
Look-ahead distance	l	0.2 m
Obstacle influence radius	d_0	1.5 m
Communication threshold	d_{th}	1.8 m
Safe distance fraction	$d_{\text{safe}}/d_{\text{th}}$	0.85
Penetration threshold	d_{pen}	0.05 m
Barrier saturation cap	F_{cap}	50.0 N
V-shape scale	—	0.6

B. Bayesian Hyperparameter Optimization

Manual tuning of APF hyperparameters often leads to fragile, overfitted controllers. To guarantee generalizability, the parameter vector $\mathbf{x} = [k_{\text{form}}, k_{\text{att}}, k_{\text{rep}}, k_{\text{vortex}}, k_{\text{conn}}]$ was optimized using Bayesian inference over a Gaussian process surrogate model [1].

The optimization environment consisted of a multi-scenario gauntlet: a narrow staggered-gate wall, a two-pillar corridor, an S-curve dogleg forcing agents into physical map boundaries, and a dense gate field. The scalar cost function J heavily penalized obstacle and wall collisions ($J_{\text{penalty}} = 50{,}000$) and network disconnections ($J_{\text{disconnect}} = 500{,}000$), while rewarding rapid goal acquisition through a cumulative time-weighted formation error integral. The surrogate model converged within 60 evaluations, identifying the optimal gains listed in Table II.

TABLE II
BAYESIAN-OPTIMIZED APF GAIN PARAMETERS

k_{form}	k_{att}	k_{rep}	k_{vortex}	k_{conn}
26.327	32.240	2.100	0.214	17.679

VII. Simulation Results and Discussion

A. Scenario Descriptions

Two scenarios were evaluated. *Scenario I: Staggered Gate Field* (Fig. 1) uses a fixed layout of eight circular obstacles ($r = 0.8$ m) aligned at $x = 0$ in the workspace $[-8, 8] \times [-5, 5]$ m, creating a narrow central passage. The swarm navigates from $(-6, 0)$ m to $(6, 0)$ m. *Scenario II: Random Forest Gauntlet* (Fig. 2) deploys 32 randomly clustered obstacles of mixed radii within $[-6, 6] \times [-6, 6]$ m on the larger workspace $[-12, 12] \times [-8, 8]$ m, with the swarm traversing from $(-10, 0)$ m to $(10, 0)$ m.

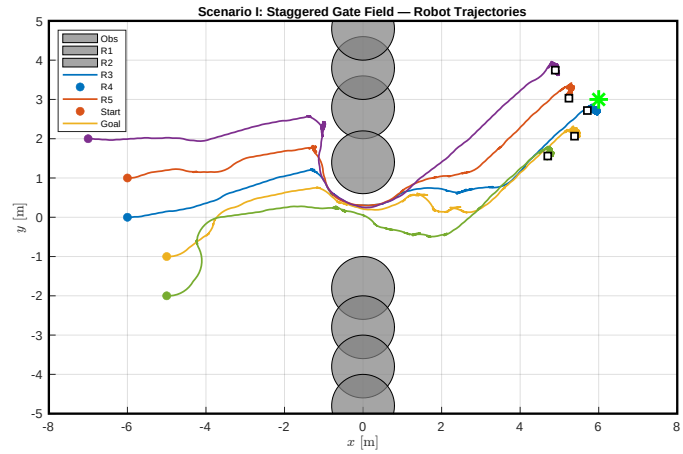

Fig. 1. Scenario I trajectory: Staggered Gate Field. The formation squeezes through the central gap ($x = 0$) and re-establishes the V-shape beyond the obstacles.

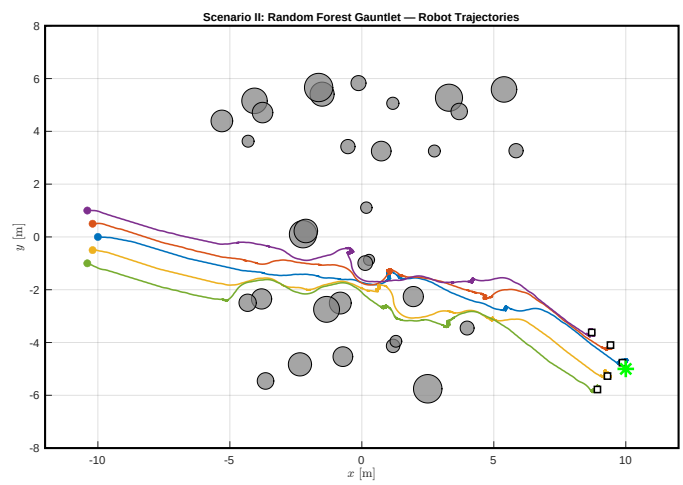

Fig. 2. Scenario II trajectory: Random Forest Gauntlet. The swarm navigates 32 randomly placed obstacles across the full 90 s simulation horizon while maintaining $\lambda_2 > 0$ and formation error below 1.03 m peak.

B. Formation Error Convergence

Fig. 3 shows the formation error norm $\|e(t)\|$ for both scenarios. In Scenario I, the error begins at approximately $\|e(0)\| \approx 1.34$ m as the swarm starts in a loose cluster and converges rapidly as the distributed attractive forces pull agents toward their V-shape targets. A pronounced spike is observed as the formation passes through the staggered gate at $x = 0$: the dynamic squeeze Eq. (17) temporarily reduces $k_{\text{form},i}$ to 10% of baseline, allowing the outer wing agents to decouple from the rigid geometry. The error stabilizes at approximately 0.19 m over the remainder of the 60 s simulation horizon. Scenario II exhibits a peak error of 1.02 m and a final error of 0.22 m, with multiple transient spikes corresponding to individual obstacle clusters and sustained convergence in the open regions between them. In both cases the error is non-zero but bounded during transit, confirming that the squeeze mechanism successfully resolves the deadlock without abandoning the formation entirely.

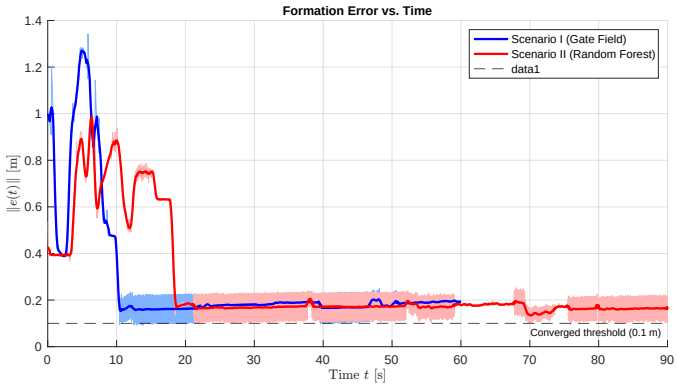

Fig. 3. Formation error $\|e(t)\|$ over time. Peak errors of 1.34 m (Scenario I) and 1.02 m (Scenario II) occur during obstacle passages and reflect the dynamic formation squeeze activating; the error stabilizes below 0.22 m in the open field.

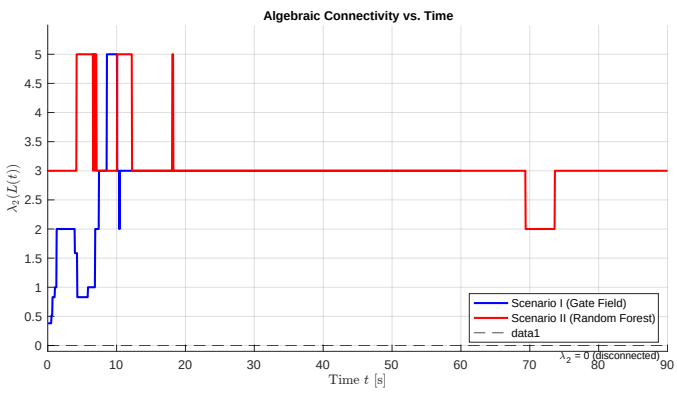

Fig. 4. Algebraic connectivity $\lambda_2(L(t))$. Both scenarios maintain $\lambda_2 > 0$ throughout (minimum values: 0.38 in Scenario I, 2.00 in Scenario II), confirming uninterrupted network connectivity.

C. Connectivity Preservation

Fig. 4 plots $\lambda_2(L(t))$ for both scenarios. The algebraic connectivity remains strictly positive throughout both trials, confirming that no communication link was ever severed. In Scenario I, λ_2 exhibits a shallow valley as agents compress through the central gate, a direct consequence of agents temporarily approaching $d_{\text{th}} = 1.8$ m. The saturated tangent barrier Eq. (18) activates only during this brief window (when $d_{ij} > d_{\text{safe}}$), applying a restoring force that prevents any edge from reaching d_{th} and thus keeps λ_2 bounded away from zero. In Scenario II, λ_2 fluctuates more broadly due to the stochastic obstacle layout, but the minimum recorded values of $\lambda_2^{\min} = 0.38$ (Scenario I) and 2.00 (Scenario II) remained strictly positive, validating the connectivity guarantee.

D. Discussion

The results demonstrate that the three proposed mechanisms interact coherently. The dynamic formation squeeze prevents the topological deadlock that rigid APF formations would encounter at narrow passages. The saturated tangent barrier provides a *reactive* connectivity restoring force that incurs zero drag in the open field, unlike the continuous drag of classical rational barriers [11]. The distributed target tracking ensures that even if the network were transiently severed by an extreme obstacle cluster, individual sub-swarms would self-navigate toward their personal targets and reconnect naturally.

The high-frequency oscillation visible in the $\|e(t)\|$ plots is a numerical artefact of the *binary* adjacency model: as robots move near d_{th}, individual edges switch on and off between timesteps, changing which pairs contribute to the error norm and producing a sawtooth ripple around the true mean. This is not noise in the control sense — it does not reflect a physical force oscillation — but is rather a consequence of the hard-threshold graph topology. Replacing the binary adjacency with a smooth, distance-weighted kernel would eliminate the switching artefact at the cost of a more involved connectivity analysis.

The primary limitation of the current architecture is the reliance on a continuous-time Lyapunov argument for a discrete-time implementation. While the penetration fail-safe and barrier cap provide practical safety guarantees, a formal discrete-time Lyapunov proof would strengthen the theoretical contribution.

VIII. Conclusion

This paper presented a decentralized formation controller for five unicycle robots that simultaneously achieves V-shape formation stabilization, static obstacle avoidance, and strict communication connectivity preservation in cluttered two-dimensional environments. The nonholonomic kinematic constraint was resolved via feedback linearization through a look-ahead point, mapping each unicycle onto a fully actuated virtual integrator and enabling direct application of gradient-descent control laws. Three complementary mechanisms were introduced: a *dynamic formation squeeze* that adaptively relaxes the rigid formation geometry in narrow corridors; a *saturated tangent barrier* that incurs zero connectivity drag in the open field and activates only when an inter-agent link approaches the communication threshold; and *leaderless distributed target tracking* that enables automatic network self-recovery after transient topological splits. A Lyapunov argument demonstrated that the safe connected invariant set $\Omega = \{\mathbf{p}_v \in \mathcal{W}_{\text{free}}^N : \lambda_2(L) > 0\}$ is forward invariant under the proposed control law, and the five APF hyperparameters were tuned via Bayesian optimization to ensure generalizability across diverse obstacle environments.

Simulations in a structured staggered gate field and a random obstacle forest confirmed successful goal acquisition with zero collisions. The algebraic connectivity remained strictly positive throughout all trials ($\lambda_2^{\min} = 0.38$ in Scenario I and 2.00 in Scenario II), validating the connectivity guarantee in practice.

Three directions are identified for future work. First, extending the framework to dynamic obstacles and moving agents is a natural next step toward real deployment. Second, replacing the binary adjacency graph with a smooth distance-weighted kernel would eliminate the edge-switching artefact observed in the formation error and enable a cleaner barrier potential formulation. Third, a formal discrete-time Lyapunov proof is needed to close the gap between the continuous-time stability argument and the Euler-integrated implementation.

REFERENCES

[1] Bobak Shahriari, Kevin Swersky, Ziyu Wang, Ryan P. Adams, and Nando de Freitas. Taking the human out of the loop: A review of Bayesian optimization. *Proceedings of the IEEE*, 104(1):148–175, 2016.

[2] Heyang Wang, Ming Yue, Xu Sun, and Xudong Zhao. Cooperative Formation Control Strategy for Multi-robot System Based on APF Algorithm and Sliding Mode Estimator. *Robotics and Autonomous Systems*, 193:105075, November 2025.

[3] Oussama Khatib. Real-time obstacle avoidance for manipulators and mobile robots. *The International Journal of Robotics Research*, 5(1):90–98, 1986.

[4] N.E. Leonard and E. Fiorelli. Virtual leaders, artificial potentials and coordinated control of groups. In *Proceedings of the 40th IEEE Conference on Decision and Control (Cat. No.01CH37228)*, volume 3, pages 2968–2973 vol.3, December 2001.

[5] Reza Olfati-Saber. Flocking for multi-agent dynamic systems: Algorithms and theory. *IEEE Transactions on Automatic Control*, 51(3):401–420, March 2006.

[6] Zhen Kan, Ashwin P. Dani, John M. Shea, and Warren E. Dixon. Network Connectivity Preserving Formation Stabilization and Obstacle Avoidance via a Decentralized Controller. *IEEE Transactions on Automatic Control*, 57(7):1827–1832, July 2012.

[7] Elon Rimon and Daniel E. Koditschek. Exact robot navigation using artificial potential functions. *IEEE Transactions on Robotics and Automation*, 8(5):501–518, October 1992.

[8] S. S. Ge and Y. J. Cui. New potential functions for mobile robot path planning. *IEEE Transactions on Robotics and Automation*, 16(5):615–620, October 2000.

[9] Meng Ji and Magnus Egerstedt. Distributed Coordination Control of Multiagent Systems While Preserving Connectedness. *IEEE Transactions on Robotics*, 23(4):693–703, August 2007.

[10] D.V. Dimarogonas and K.H. Johansson. Bounded control of network connectivity in multi-agent systems. *IET Control Theory & Applications*, 4(8):1330–1338, August 2010.

[11] Dimitra Panagou, Dušan M. Stipanović, and Petros G. Voulgaris. Distributed Coordination Control for Multi-Robot Networks Using Lyapunov-Like Barrier Functions. *IEEE Transactions on Automatic Control*, 61(3):617–632, March 2016.

[12] K. D. Do. Formation tracking control of unicycle-type mobile robots with limited sensing ranges. *IEEE Transactions on Control Systems Technology*, 16(3):527–538, May 2008.

[13] Giuseppe Oriolo, Alessandro De Luca, and Marilena Vendittelli. WMR control via dynamic feedback linearization: Design, implementation, and experimental validation. *IEEE Transactions on Control Systems Technology*, 10(6):835–852, November 2002.